
SOFTWARE VALIDATION & VERSION HISTORY REPORT

PROJECT : *CalcSuite Pro: Digi-ToolKit*
AUTHOR : *Yousuf S. R. Sakka*
DATE : *24 April 2026*
STATUS : *Final Release (v7.6)*

1. PROJECT OVERVIEW

CalcSuite Pro is a multi-tool mathematical workstation featuring secure Abstract Syntax Tree (AST) evaluation, Cramer's Rule matrices, and a custom polynomial dictionary engine for natural-language algebra parsing.

2. MASTER TEST CASES & VALIDATION SUITE

The following cases were executed to validate the engine's logical accuracy and error handling.

TAB 1: SCIENTIFIC CALCULATOR

- **Standard Arithmetic:** $2 * (5 + 5)$
 - **Expected Result:** 20
- **Implicit Multiplier:** 2π
 - **Expected Result:** 6.2831853
- **Trigonometry (DEG):** $\sin(90)$
 - **Expected Result:** 1.0
- **Advanced Logic:** $\text{FIBS}(10)$
 - **Expected Result:** Popup window displaying 0, 1, 1, 2, 3, 5, 8, 13, 21, 34
- **Syntax Guard:** $2x + 5$
 - **Expected Result:** 450ms human-speed delay -> Redirects to Algebra Mode.

TAB 2: SOLVERS

- **Quadratic (Real):** $a=1, b=5, c=6$
 - **Expected Result:** $x_1 = -2, x_2 = -3$
- **Quadratic (Complex):** $a=1, b=2, c=5$
 - **Expected Result:** $x_1 = -1 + 2i, x_2 = -1 - 2i$
- **Linear Fallback:** $a=0, b=2, c=-10$
 - **Expected Result:** Triggers 'Linear Fallback' note $\rightarrow x = 5$
- **Linear 2x2 (Unique):** EQ I: 2,3,10 | EQ II: 4,-1,6
 - **Expected Result:** $x = 2, y = 2$
- **Linear (No Solution):** EQ I: 2,4,10 | EQ II: 2,4,20
 - **Expected Result:** Triggers 'Parallel Lines / Inconsistent' system guard.

TAB 3: SMART ALGEBRA

- **Numeric Bypass:** $5 + 10 * 2 = 25$
 - **Expected Result:** Note: Mathematical Statement is True. LHS: 25, RHS: 25
- **Simplification:** $2x + 5x - x$
 - **Expected Result:** Simplified: $6x$
- **Linear (1 Var):** $3x - 5 = 10$
 - **Expected Result:** Rearranged: $3x - 15 = 0 \rightarrow x = 5$
- **Linear (2 Var):** $2a - 6b = 19, 4a + 4b = 12$
 - **Expected Result:** $a = 4.625, b = -1.625$
- **Bracket Expansion:** $(x-2)(x+5)(x-1)(x+1) = 6$
 - **Expected Result:** Polynomial iteration $\rightarrow$ Rearranged $\rightarrow +4 = 0$
- **Polynomial Guard:** $x^3 + 2x = 0$
 - **Expected Result:** Triggers Degree-3 analytical limitation note.
- **Cross-Term Guard:** $2xy + 5 = 0$
 - **Expected Result:** Triggers multi-variable cross-term limitation note.
- **Underdetermined System:** $7a+8b-9a+12b = 12a-13b+5a-8$
 - **Expected Result:** Engine rearranges to $-19a+33b+8 = 0 \rightarrow$ Triggers 'Underdetermined' guard $\rightarrow$ Instructs user to provide a second equation.
- **Format Guard (Non-Linear):** $(x-3)(x+4) = 2x$
 - **Expected Result:** Triggers 'Format Error' $\rightarrow$ Prompts user to move all variables to the left side to enforce standard polynomial form.
- **Format Guard (Quadratic):** $x^2 + 4x = 2x + 7$
 - **Expected Result:** Triggers 'Format Error' $\rightarrow$ Prompts user to move all variables to the left side.

TAB 4: HEALTH / BMI

- **Metric System:** Age 25, Male, 100kg, 180cm
 - **Expected Result:** BMI: 30.9 (Obese) + Athlete context note.
- **Imperial System:** Age 25, Female, 140lb, 5ft 5in
 - **Expected Result:** BMI: 23.3 (Normal) + Female body-fat context note.
- **Age Guard:** Age 15, Male, 60kg, 160cm
 - **Expected Result:** Calculates BMI but overrides category -> requires chart.

TAB 5: UNIT CONVERTER

- **Standard Conversion:** Length -> 1 Metre to Centimetre
 - **Expected Result:** 100 cm
- **Advanced Conversion:** Temperature -> 0 Celsius to Fahrenheit
 - **Expected Result:** 32 °F
- **Swap Functionality:** Press 'Swap' on previous result
 - **Expected Result:** Swaps to 32 Fahrenheit -> 0 Celsius.

3. EXTENDED VERSION HISTORY

This application underwent continuous daily iteration to refine security, logic handling, and user interface responsiveness.

Version	Date	Status	Key Upgrades & Architectural Changes
v1.0	30-03-2026	Legacy	Initial deployment. Basic arithmetic GUI logic established using standard libraries.
v1.2	01-04-2026	Legacy	Incremental logic optimizations and operational bug fixes.

v2.0	02-04-2026	Legacy	Major UI restructuring and visual layout adjustments.
v3.2	03-04-2026	Legacy	Multi-tool expansion. BMI and Unit Converter algorithms integrated.
v3.3	04-04-2026	Legacy	Refinement of converter parameters and unit scaling.
v4.0	07-04-2026	Legacy	Implementation of history tracking and state persistence.
v4.2	08-04-2026	Legacy	History UI optimizations and text formatting updates.
v5.0	09-04-2026	Legacy	Security & UI Overhaul. CustomTkinter implemented. Unsafe eval() replaced by strict AST parsing.
v6.0	10-04-2026	Legacy	Core solver engine mathematical foundation established.
v6.1	11-04-2026	Legacy	Quadratic solver edge-cases and imaginary number handling addressed.
v6.2	12-04-2026	Legacy	Linear 2x2 solver logic and Cramer's Rule matrix implementation.
v6.3	13-04-2026	Legacy	Error handling and validation deployment for the Solvers tab.
v7.0	15-04-2026	Stable	Smart Algebra logic introduced. Natural language parsing for single-variable equations.
v7.2	16-04-2026	Stable	Parsing improvements for algebraic terms and string extraction.

v7.3	20-04-2026	Stable	Simplification routing and numeric bypass logic integrated.
v7.4	21-04-2026	Stable	Iterative FOIL logic added for bracket expansion.
v7.5	22-04-2026	Stable	The "Pro" Overhaul. Digi-ToolKit branding applied. Smart cross-tab redirects built.
v7.6	24-04-2026	Final	Stability Update: Dictionary-based polynomial engine implemented. Epsilon floating-point tolerance deployed. Legacy code fully pruned. UI Geometry Guard: Patched character-overflow bugs in the Scientific Display (Consolas 28 font boundary) by optimizing the FIBS exception string to exactly 20 characters. Symmetry Patch (Algebra): Enforced syntax uniformity in the Dispatcher by mapping single-bracket exponential inputs (e.g., $(x-3)^2$) to a dedicated Quadratic Guardrail rather than falling back to generic Linear limits. Parser Stability: Hardened AST regex protocols to safely strip leading zeros (e.g., 0090) to prevent fatal Python octal-conversion crashes.